Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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Abstract

Parametric Value-at-Risk and Expected Shortfall forecasts underperform in states that can be flagged in real time. A per-asset, per-day score of residual-shape stress predicts when a distribution-free tail beats a GARCH-t benchmark, an edge large in its top decile and negligible elsewhere. A jump-robust refit traces most of that edge to variance overstatement after a shock, and a smaller but significant part to residual conditional shape. The forecasts come from an amortized semiparametric estimator; its pooled quantile component transfers to newly listed assets in characteristics-only form, and it attains the lowest joint (VaR, ES) loss against GARCH-t and FHS.

Keywords: Value-at-Risk; Expected Shortfall; misspecification; quantile regression; amortized estimation.

JEL classification: C14; C52; C53; C58; G17.

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Introduction

A risk model is judged on its worst days, and those are the days on which a fixed parametric tail is most likely to be wrong. The error is costly in both directions. A forecast that runs high ties up capital against risk that never arrives, and one that runs low leaves too little against the loss that does. Deciding when a more flexible model is worth its added complexity is therefore a practical question. The question also applies outside Basel-regulated bank trading desks, including to asset managers and hedge funds that forecast conditional tail risk.

Value-at-Risk (VaR) and Expected Shortfall (ES), the forecasts at issue, summarize the downside risk of a trading book: VaR is the lower-tail quantile of the conditional return distribution, and ES the average loss beyond it.

Three model classes produce these forecasts, and they differ in how much of the conditional return distribution they fix by assumption (Kuester et al., 2006; Nieto and Ruiz, 2016). Fully parametric filters, generalized autoregressive conditional heteroskedasticity (GARCH) and its leverage and skew extensions (Bollerslev, 1986; Glosten et al., 1993), deliver conditional scale dynamics that are hard to improve within the ARCH family (Hansen and Lunde, 2005) but read the tail off an assumed innovation density. Semiparametric filters keep that scale and estimate the innovation law from the standardized residuals, empirically (Barone-Adesi et al., 1999), through an extreme-value tail (McNeil and Frey, 2000), or by modeling the quantile directly (Engle and Manganelli, 2004), and Francq and Zakoïan (2015) supply the two-step estimator's asymptotic theory. Distribution-free conditional quantile estimation by pinball-loss minimization is itself an econometric idea (Koenker and Bassett, 1978), and machine learning has scaled it up. Its members here are gradient-boosted quantile trees, neural implicit quantile networks (IQN) (Dabney et al., 2018), and the quantile-network generator of generative Bayesian computation (GBC) (Polson and Sokolov, 2023). GBC is a simulation-based posterior method. It becomes a conditional quantile learner once its forward simulator is replaced by observed state and return pairs, as Section 1 does. These learners place no distributional restriction on the innovation law, but they must learn it from data and from a chosen state vector. Machine-learning models now forecast realized volatility better than heterogeneous-autoregressive benchmarks (Bucci, 2020; Christensen et al., 2023), and when pooled across a cross-section of US stocks they transfer to names outside the training set (Zhang et al., 2024). This paper asks whether a comparable advantage exists for the quantile learners studied here, whether it reaches the regulatory tail, and on which days it appears.

Industry practice sits between these classes. Under the Basel Fundamental Review of the Trading Book (FRTB), the workhorse internal models are historical simulation (HS), by far the most common method in bank disclosures (Pérignon and Smith, 2010), and its GARCH-filtered variant, filtered historical simulation (FHS) (Barone-Adesi et al., 1999). The regulatory choice is not between a fixed model and a proprietary one but between a prescribed standardized formula and a bank's own internal model, and many banks have moved back toward the standardized approach as internal models became costly to maintain. The buy side runs the same classical toolkit: asset managers and hedge funds, though outside Basel, forecast VaR and ES with historical simulation, parametric filters, and Monte Carlo, largely through vendor risk systems. The comparison in this paper is therefore against the methods in standard use across the industry.

Forecasts are judged by VaR exception counts at 99% and 97.5%, and the reported measure is Expected Shortfall at the 97.5% level. Reported ES enters the market-risk capital a bank holds against its book (Basel Committee on Banking Supervision, 2019), so errors are costly in both directions. A forecast that runs high ties up capital that could fund positions, while one that runs low raises the multiplier on the charge as exceptions accumulate. A method that claims to beat the industry standard must therefore beat HS and FHS on a jointly consistent $(\mathrm{VaR}, \mathrm{ES})$ score at the 97.5% level (Fissler and Ziegel, 2016), since ES alone cannot be scored (Gneiting, 2011), and it must do so under exception-test discipline (Nolde and Ziegel, 2017). Beating a vanilla GARCH on average loss is therefore neither sufficient nor trivial, because bank internal models have historically failed to beat a simple GARCH fitted to their own profit and loss (Berkowitz and O'Brien, 2002).

The advantage of the flexible model over the parametric benchmark is concentrated in particular states. On the roughly ninety percent of asset-days (one name on one trading date) when residual-shape stress is low, the estimator we use and the parametric benchmark are statistically indistinguishable on forecast accuracy, so holding the flexible model carries no detectable penalty (Section 4 reports the intervals). The gains are concentrated in the minority of days when residual-shape stress is high. We ask whether an observable signal available at the forecast date can identify those asset-days in advance. It is a conditional predictive-ability question in the sense of Giacomini and White (2006): does a signal known at the forecast date predict which of two forecasts will be more accurate on that date? We construct such a signal, a real-time misspecification score, and show that it orders the flexible model's advantage. We then build a risk model that exploits the ordering while matching the benchmark elsewhere. Because its residual-shape model is pooled across names and conditioned on characteristics, it also delivers forecasts for assets that have no return history at all, something no per-asset model, parametric or not, can do. Pooled estimation is known to improve volatility forecasts out of sample (Bollerslev et al., 2018; Zhang et al., 2024); here it is carried to the day-one limit and into the tail.

The score is built from the oldest specification diagnostics for a GARCH filter, the excess kurtosis and asymmetry of its standardized residuals (Bollerslev, 1987; Hansen, 1994). It is computed on a trailing window so that it is available in real time, in the spirit of the density-forecast diagnostics of Berkowitz (2001). We use these recent-residual diagnostics to predict when distribution-free quantile methods improve on parametric VaR and ES. Where residual-shape stress is high the flexible estimator has a large and significant advantage, and where it is low the advantage is statistically indistinguishable from zero in the design panel and a small positive in the point-in-time universe, with no detectable loss in either. The score orders the magnitude of the advantage rather than defining a sharp threshold: the edge is concentrated in the top decile but does not vanish below it.

The paper's main contribution is the misspecification frontier: a real-time, per-asset, per-day score that predicts when distribution-free quantiles beat a GARCH-t benchmark, with significance from Diebold–Mariano tests on date-averaged loss differentials (the per-date DM of Section 3). The same kurtosis–edge gradient is traced across four universes: CRSP US equities (top-decile edge +2.98%, DM 10.5), foreign exchange, where the decisive standalone wins are confined to the hyperinflation tier (pooled DM 4.12), twenty-six country indices (a developed-to-frontier gradient), and forty-three cross-asset instruments.

A jump-robust decomposition then identifies what drives that advantage. Measured against a jump-robust GARCH whose variance response to a large shock is bounded, we show that most of it is the standard filter's post-outlier scale error, with a smaller but significant component surviving a robust scale (the top-decile +2.98% becomes +0.3–0.5%, DM 2.5–4.7). On large caps with intraday data, re-estimation on realized-volatility residuals leaves the score within noise. This attribution, drawn on a 200-name panel and tied to a live score, is, to our knowledge, new to the flexible-versus-parametric tail-risk literature. The Korea-2026 crash, which unfolded under circuit breakers and left residual kurtosis in its ordinary range, is the negative control that sharpens the claim: a price crash is not residual misspecification, and GARCH-t wins there, as the frontier predicts.

The pooled shape model also transfers across assets: a single fit across hundreds of names (amortized, in the machine-learning sense that one fit serves every asset) carries transferable information about conditional tail shape: it replaces per-asset estimation of the innovation shape and transfers to names never seen in training. Its gains over own-history benchmarks are largest in the first trading month, at roughly 6–10% of pinball loss. In its characteristics-only form it forecasts VaR and ES for assets with no return history at all, the day-one regime in which no per-asset model exists.

The frontier is implemented with a semiparametric estimator that follows the two-step logic of filtered historical simulation and GARCH with an extreme-value-theory (EVT) tail (Barone-Adesi et al., 1999; McNeil and Frey, 2000). It keeps the parametric filter's conditional scale, the component the ARCH family already estimates well once a leverage term is allowed (Hansen and Lunde, 2005), and replaces the innovation law where it fails. The forecast is therefore a GARCH conditional scale times a state-conditioned nonparametric residual quantile, with an extreme-value left tail and an optional split-conformal overlay. The resulting estimator and CAViaR (Engle and Manganelli, 2004) both remain in the 90% Model Confidence Set (MCS) (Hansen et al., 2011), and it improves significantly on GARCH-t, FHS, EWMA, HS, and GJR-skew-t. Its EVT-tailed accuracy layer attains the lowest joint $(\mathrm{VaR}, \mathrm{ES})$ score, the Fissler–Ziegler zero-homogeneous (FZ0) loss of Fissler and Ziegel (2016) and Patton et al. (2019), against the daily benchmarks in standard use, GARCH-t and FHS, at both FRTB levels. That ranking survives annual refitting, with exception counts near nominal through a 2008-crisis window. It also keeps the lowest FZ0 against the FZ-estimated dynamic ES models of Patton et al. (2019) and Taylor (2019), each estimated per name by FZ0 minimization, winning at 1% and tying Taylor at 2.5%. Two results motivate the score without supplying its mechanism. The first is the standard identity that excess pinball loss is locally quadratic in the quantile wedge, scaled by the local density (Gneiting, 2011; Ehm et al., 2016; Steinwart and Christmann, 2011), so that a wedge at a level generates regret there. The second, a tail-wedge proposition, shows how a misspecified innovation shape opens such a wedge in the tail even when the scale is right (McNeil and Frey, 2000; Francq and Zakoïan, 2015). The decomposition indicates that the score primarily captures scale error, while the surviving shape component is smaller.

The loss profile is asymmetric in the score. When the frontier is quiet the estimator ties or edges the parametric benchmark (calm-quintile edge ≈ 0). In the top decile the improvement is +2.98%, and on the hyperinflation and capital-control currencies, where the parametric assumption breaks entirely, it is 12–14%. These are reductions in a consistent scoring function for the quantile (Gneiting, 2011), not a monetary welfare measure. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level. Because the ranking of misspecified forecasts can depend on which consistent scoring function is used (Patton, 2020), the joint FZ0 score of Section 5 serves both as the regulatory comparison and as a check on that ordering under a second score.

The paper proceeds as follows. Section 1 places the method against the industry benchmark and the forecasting literature. Section 2 develops the semiparametric estimator and the two results that motivate the frontier. Section 3 sets out the data and the forecast-evaluation protocol. Section 4 presents the misspecification frontier, and Section 5 reports the joint VaR and ES evaluation. Section 6 covers cross-sectional transfer, longer horizons, and cross-asset evidence, and Section 7 concludes.

1 Related literature and benchmark models

1.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail). The return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$ for continuous F_t. In general $e_t^\alpha = \alpha^{-1}\int_0^\alpha F_t^{-1}(v)\,dv$, the coherent form used throughout (Artzner et al., 1999; Acerbi and Tasche, 2002). The regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 2.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at a stressed $\text{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket), while backtesting remains VaR-exception based. More than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a trading desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint. It requires accuracy on the joint (VaR, ES) score at 97.5% (Fissler and Ziegel, 2016; Patton et al., 2019), since ES alone cannot be scored (Gneiting, 2011), and on pinball, together with the unconditional-coverage test of Kupiec (1995) and the conditional-coverage test of Christoffersen (1998), the joint likelihood-ratio test of correct coverage and breach independence, at both levels. Significance is assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011). Real-time monitoring of VaR and ES forecast adequacy is developed by Hoga and Demetrescu (2023); the score of Section 2.6 is a cross-sectional, per-asset monitor of the same forecasts, used to rank states rather than to signal a structural break.

1.2 The parametric family, the simulation family, and CAViaR

The comparison set spans the models banks run. It includes historical simulation, exponentially weighted moving average (EWMA) variance with the RiskMetrics decay 0.94 (J.P. Morgan/Reuters, 1996) (the long-memory successor in Zumbach, 2007 is not run here), GARCH(1,1)-t (Bollerslev, 1986, 1987), and the GJR (Glosten–Jagannathan–Runkle) GARCH-skew-t (Glosten et al., 1993; Hansen, 1994), which adds a leverage term and a skewed innovation density. It also includes GARCH-FHS (Barone-Adesi et al., 1999; Hull and White, 1998), which pairs a parametric scale with empirical standardized residuals and is the filtered method that the comparison literature ranks closest to the top (Kuester et al., 2006; Nieto and Ruiz, 2016). The strongest non-nested semiparametric rival in the set is SAV-CAViaR (conditional autoregressive value-at-risk) (Engle and Manganelli, 2004), to which the comparison literature is less favorable at the 1% level (Kuester et al., 2006). The SAV (symmetric absolute value) specification models the quantile directly by regression quantiles (Koenker and Bassett, 1978), $q_t(\tau) = b_0 + b_1\, q_{t-1}(\tau) + b_2\, |r_{t-1}|$ for a quantile level τ (defined in Section 2.1), the quantile analogue of an absolute-value GARCH recursion (the indirect-GARCH CAViaR is the analogue of the squared recursion). The comparison set of Table 3 does not include standalone GARCH-EVT (McNeil and Frey, 2000), which Kuester et al. (2006) rank first among classical methods. Its filter-then-tail logic is absorbed into the estimator's own EVT stage (Section 2), so the EVT-tailed layer is its analogue within the comparison set. A same-rows comparison against standalone McNeil–Frey GARCH-EVT, per name and with a pooled tail, is reported in Sections 4 and 5 and in the Online Appendix: the conventional EVT tail carries no frontier of its own, and the estimator beats it by 2.64% (DM 8.9) in the top kurtosis decile and on the joint score at both levels.

1.3 Distribution-free conditional quantiles and GBC

For the pooled, cross-sectional design the closest antecedent is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names. That transfer is evidence of a shared volatility mechanism, which our amortized design also exploits. For the VaR comparison the antecedent is Kuester et al. (2006), and for machine-learning volatility forecasting Bucci (2020) and Christensen et al. (2023). Deep neural-network quantile regression applied directly to VaR is the recent comparison closest to our nonparametric stage (Chronopoulos et al., 2024). Two further semiparametric VaR and ES papers in this journal are direct precedents: Jiang et al. (2022) estimate a single-index expectile model, and Luo et al. (2025) select low-frequency predictors for joint tail forecasts through a GARCH-MIDAS design with adaptive-Lasso selection. Our object differs from both: rather than proposing a new model or selecting predictors, we ask when an observable signal identifies the states in which flexible tail forecasts beat parametric ones, and we decompose the source of that conditional advantage. Relative to the pooled-volatility literature, our object is state dependence in conditional tail-risk forecast performance rather than average realized-variance forecast performance, with the regulatory tail held to calibration, exception-test, and capital discipline. The question here is where a flexible model helps at all: on roughly ninety percent of asset-days it does not improve on the parametric benchmark, which cautions against extrapolating average machine-learning superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at each listing age. On tabular state, gradient-boosted trees outperform the implicit quantile network, the reverse of the intraday ranking in Zhang et al. (2024), where neural networks dominate tree-based models. At their daily horizon that dominance already disappears, and our target is a residual quantile curve rather than realized variance, with a different feature set.

The nonparametric family estimates $Q_\phi(\tau \mid s_t)$ directly by pinball-loss empirical risk minimization (ERM) (Koenker and Bassett, 1978; Steinwart and Christmann, 2011). Its two members here are gradient-boosted quantile trees (GBM) (Friedman, 2001; Meinshausen, 2006) and the implicit quantile network of Dabney et al. (2018), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). In the IQN the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time.

Read as ERM on observed pairs, GBC's quantile generator coincides with a conditional quantile network trained on data (Dabney et al., 2018; Koenker and Bassett, 1978): the forward simulator is not needed, and the same object applies to observational time series (Section 2.5). The GBM is the shape model used throughout, and the IQN is retained as a benchmark. Its generative reading (draw $\tau \sim U(0, 1)$, output $Q_\phi(\tau \mid s)$) is a capability we note without using, since sampling and posterior uncertainty are outside this paper's scope. Pinball ERM can be read as the mode of a Gibbs posterior built from the loss in place of a likelihood (Jiang and Tanner, 2008; Bissiri et al., 2016); no result below uses that posterior. Distribution-free finite-sample coverage comes from split-conformal prediction (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019), which shifts a predicted quantile by an order statistic of held-out calibration errors so that marginal coverage holds under exchangeability with no distributional model. Adaptive conformal inference (Gibbs and Candès, 2021) replaces the fixed shift by an online update whose long-run coverage holds under arbitrary distribution shift, at the cost of the finite-sample guarantee. Barber et al. (2023) bound the coverage loss when exchangeability fails, the case for time-ordered residuals.

2 Semiparametric VaR and ES model

Each estimator below can be reimplemented from the formulas and algorithm boxes given. A glossary of acronyms opens the Online Appendix.

2.1 Notation and objects

Let r_t be the return of an asset on day t and $\mathcal{F}_{t-1}$ the information available before t. The conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$ is $Q_t(\tau) = \inf\{q : \Pr(r_t \leq q \mid \mathcal{F}_{t-1}) \geq \tau\}$, the number the return falls below with probability τ. Throughout, $\tau \in (0, 1)$ is a generic quantile level and $\alpha \in \{0.01, 0.025\}$ denotes the reported regulatory levels; both index the same object. One sign convention is used throughout. The paper works in *return space*, with $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1} \int_0^\tau Q_t(v)\, dv \leq q_\tau$, the coherent form of Expected Shortfall (Artzner et al., 1999; Acerbi and Tasche, 2002). This is the convention of the FZ0 scoring function $(v, e < 0)$ (Fissler and Ziegel, 2016; Patton et al., 2019) and of each table, where ES values such as -6.37 are return-space conditional means. The regulatory *loss-space* numbers are their negatives, $\mathrm{VaR}_\tau^{\mathrm{loss}} = -q_\tau$ and $\mathrm{ES}_\tau^{\mathrm{loss}} = -e_\tau$, used only when quoting FRTB levels. The two are never mixed in one expression.

The *standardized residual* $z_t = (r_t - \mu_t)/\sigma_t$ is the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law. The state vector s_t collects trailing realized volatility, the misspecification scores of Section 2.6, and lags, and is what the nonparametric quantiles condition on. A quantile level $\tau \sim \lambda$ is drawn from a base distribution λ. The learned generator $H_\phi(s, \tau)$ maps (state, level) to a quantile in the sense of Polson and Sokolov (2023): $z \stackrel{D}{=} H(s, \tau)$ with $\tau \sim U(0, 1)$ is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(x) \;=\; x\,(\tau - \mathbb{I}\{x < 0\}), \qquad x = z - q, \tag{1}$$

whose expectation is minimized by a conditional τ-quantile (Koenker and Bassett, 1978; Gneiting, 2011)—uniquely when that quantile is unique—the way squared error elicits the mean. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below). The loss is density-free, so each estimator below dispenses with a likelihood.

Choice of scoring function. We score by pinball, a consistent scoring function for the quantile (Gneiting, 2011), at the decision-relevant levels rather than by a global proper scoring rule, for three reasons. (a) The continuous ranked probability score (CRPS), the proper scoring rule for the whole distribution (Gneiting and Raftery, 2007), is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\,d\tau$ (Gneiting and Ranjan, 2011). Scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. The quantile-weighted CRPS of Gneiting and Ranjan (2011) reweights the levels for this reason, and scoring at fixed levels is its point-mass case. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail, and CRPS would average those differences away against a body where all competitors agree. For the same reason, training uses tail-weighted τ-sampling. (b) Expected Shortfall is not elicitable on its own (Gneiting, 2011) (no loss function is minimized by the true ES alone), but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016), whose zero-homogeneous member, the FZ0 loss of Patton et al. (2019), is the one Section 5 uses. The FZ-scored re-evaluation of the comparison set is an extension specified in advance (Section 5). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them. Each benchmark is estimated by its own conventional criterion: Student-t or skew-t likelihood for the GARCH family, empirical residual quantiles for HS and FHS, FZ0 for the two dynamic-ES models of Section 5, and pinball for CAViaR and for the shape learners. All are then scored out of sample by the same pinball and FZ0 losses, so a model fitted on the evaluation loss carries an in-sample advantage that the out-of-sample comparison tests.

2.2 The final model: the residual-hybrid estimator

The paragraphs below define each stage of the estimator; a schematic of the full pipeline is in the Online Appendix.

The estimator is a four-stage composition in the two-step tradition of filtered historical simulation and GARCH-EVT (Barone-Adesi et al., 1999; Hull and White, 1998; McNeil and Frey, 2000). The parametric filter estimates conditional scale, which it does well and with few parameters (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005), and the nonparametric stage estimates the state-dependent shape of the innovation law, which the fixed-shape GARCH-t benchmark fixes by assumption. The parametric route to a state-dependent shape is the autoregressive conditional density model of Hansen (1994), and Stage 2 is its distribution-free counterpart. The extreme-value tail extrapolates beyond the estimation range using the generalized Pareto approximation of Stage 3 below. The conformal stage corrects the level of the quantile curve using its measured miss rate on a held-out split, with the exchangeability-based guarantee of Proposition 1.

Motivation for the design. Conditional scale is identified by a few parameters and estimated with low variance by a parametric filter, so the shape stage is where flexible methods can add value. Per-asset tail data is scarce, which motivates pooling the standardized residuals across names. The EVT stage extrapolates the tail by the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), and the conformal stage is motivated by the split-conformal guarantee of Proposition 1, exact under exchangeability and approximate for the time-ordered residuals used here, with a coverage gap bounded by the average total-variation distance between the residual vector and its swapped versions (Barber et al., 2023). Whether the four stages together improve on the benchmark out of sample is the question of Sections 4 and 5.

Stage 1 (parametric scale). Fit GARCH-t (Bollerslev, 1986, 1987),

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega_G + \alpha_G (r_{t-1} - \mu)^2 + \beta_G \sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by Student-t quasi-maximum likelihood on the estimation window of Section 3 (annually refit in ongoing use), and keep $\widehat{\mu}$, $\widehat{\sigma}_t$, and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$. The t_ν likelihood is used to estimate conditional scale; Stages 2 and 3 re-estimate the residual shape. This two-step design, a parametric filter followed by a residual quantile, is the one whose asymptotics Francq and Zakoïan (2015) derive, and McNeil and Frey (2000) fit the same filter by Gaussian pseudo-likelihood while explicitly declining to believe the innovation law it assumes.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). In the sparsely observed lower tail, where (3) is fitted on few exceedances, use the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Balkema and de Haan, 1974; Pickands, 1975; McNeil and Frey, 2000). The finite tail mean of the generalized Pareto distribution (GPD) requires $\xi < 1$, a condition each fitted tail in this paper satisfies by a wide margin.[1] On losses $X = -\widehat{z}$, set the threshold u at the empirical $(1 - p_0)$ quantile of the pooled training residual losses, with $p_0 = 0.025$. Fit the GPD $G_{\xi,\beta}(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$ to the excesses $y = X - u > 0$ by maximum likelihood with location fixed at zero, strictly on past data, giving $(\widehat{\xi}, \widehat{\beta})$; (4) is evaluated at these estimates and hats are suppressed below. The shape ξ sets how heavy the tail is and the scale $\beta > 0$ how wide; this β is the GPD scale, distinct from the GARCH persistence β_G. One (ξ, β) pair is fitted per panel and held fixed across states and levels within a fit. For $\tau \le p_0$, set

$$\widehat{q}^{\,\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\,\mathrm{EVT}}(p_0) = -u$. The formula requires $1 + \xi y/\beta > 0$, and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit. The threshold u is unconditional with respect to the state, while the Stage-2 body quantile at p_0 is the state-conditional $\widehat{q}_\phi(p_0 \mid s)$, so the two branches need not meet at the join. In the tail the forecast takes the pointwise minimum of the two, the more conservative (more negative) branch, so the pooled GPD extrapolation can tighten a thin state-conditioned body on a stressed day but never loosen it. The curve is then made non-crossing by monotone rearrangement, which cannot move it further from the true quantile curve in any L^p norm (Chernozhukov et al., 2010). Both steps are written out in (5).

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $\varepsilon_i = \widehat{z}_i - \widehat{q}_\phi(\tau \mid s_i)$ against the Stage-2 body. Let c_τ be the $\lceil (n+1)\tau \rceil$-th order statistic of $\{\varepsilon_i\}$. The finished curve is shifted per level by this scalar, $\widetilde{Q}^z_t(\tau) = Q^z_t(\tau) + c_\tau$ with Q^z_t the envelope of (5) (Vovk et al., 2005; Lei et al., 2018), the one-sided form of conformalized quantile regression (Romano et al., 2019). The shift is the last operation, after the minimum and the rearrangement. The *accuracy layer* that carries the FZ0 comparisons of Section 5 and Table 3 sets $c_\tau \equiv 0$; the *conformal overlay* applies the shift at the 97.5% level. This distribution-free layer repairs the 97.5% level that each model in the comparison set otherwise fails. The coverage statement it inherits is standard (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019) and is restated here in the form the shift uses.

Proposition 1 (split-conformal validity under exchangeability (standard))**.** *If the calibration and test errors $\varepsilon_1, \ldots, \varepsilon_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \le n$ the shifted quantile satisfies*

$$\tau \;\le\; \Pr\left(z_{n+1} \le \widehat{q}_\phi(\tau \mid s_{n+1}) + c_\tau\right) \;=\; \frac{\lceil (n+1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of ε_{n+1} among $\{\varepsilon_1, \ldots, \varepsilon_{n+1}\}$ is uniform on $\{1, \ldots, n+1\}$. The event $z_{n+1} \le \widehat{q}_\phi(\tau \mid s_{n+1}) + c_\tau$ equals $\{\varepsilon_{n+1} \le \varepsilon_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \le \lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument is in the Online Appendix, together with the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1) and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$. The guarantee is marginal over the state s_{n+1} rather than state-conditional. □

The proposition's scope is narrower than its use. In practice the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable. The proposition therefore motivates the construction. The evidence that it works here is empirical: in the exception-test repairs of Section 5, the conformal stage is the only reason any entrant passes Kupiec at 97.5%. The guarantee also attaches to the shifted body quantile alone. The finished forecast adds the same scalar to the envelope of (5), whose tail nodes take the minimum of body and EVT branches before rearrangement, so the proposition covers the shifted body while the reported curve is the shifted envelope. The exact finite-sample coverage identity need not survive that composition, so the coverage claim at the reported levels rests on these exception tests rather than on Proposition 1. Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018; Barber et al., 2023; Gibbs and Candès, 2021), but we prefer the simple split form plus an empirical audit.

One further implementation gap: the first-stage GARCH is fit on the full pre-test history, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer comparisons never read the calibration split and share their filter with the benchmark, so their information sets match. Only the overlay's shift uses the split. Re-estimating with the filter stopped at the calibration boundary leaves the accuracy advantage intact (DM 5.1 and 5.4). The strict estimator's gap to the full-window benchmark (-0.6, -2.5) matches that benchmark's own margin over its short-window twin (2.8, 3.8), so the gap reflects a quarter less estimation data rather than leakage. The static shift widens (per-name pass 34% against 48%), consistent with its disclosed conservatism. Under the same strict split the adaptive conformal update of Gibbs and Candès (2021) walks the shift from -0.51 to -0.20, lifts the per-name pass rate to 57%, and shrinks the joint-score gap to GARCH-t from DM -3.2 to -1.5 (not significant at 5%). The estimator's annual-refit schedule sits between the static and adaptive poles, and the exception tests decide between them.

The full forecast composes the stages:

$$Q_t^z(\tau) \;=\; \mathrm{R}\Big[\,\min\{\widehat{q}_\phi(\tau \mid s_t),\, \widehat{q}^{\,\mathrm{EVT}}(\tau)\}\,\mathbb{I}\{\tau \le p_0\} + \widehat{q}_\phi(\tau \mid s_t)\,\mathbb{I}\{\tau > p_0\}\Big], \qquad \widehat{Q}_t(\tau) \;=\; \widehat{\mu}+\widehat{\sigma}_t\big(Q_t^z(\tau)+c_\tau\big), \tag{5}$$

where $\mathrm{R}[\cdot]$ is the monotone rearrangement across levels (Chernozhukov et al., 2010) and $c_\tau \equiv 0$ in the accuracy layer. Q_t^z is the standardized-residual quantile curve, the single object from which both risk measures are read; the conditional scale $\widehat{\sigma}_t$ carries the day's volatility from the GARCH stage. The return-space forecasts are $\widehat{q}_{\tau,t} = \widehat{Q}_t(\tau)$ and $\widehat{e}_{\tau,t} = \widehat{\mu} + \widehat{\sigma}_t\,\big(\tau^{-1}\!\int_0^\tau Q_t^z(v)\,dv + c_\tau\big)$, the latter a numerical integral of the same curve. The body branch is the minimum on 38% of tail nodes. Computing ES_α as the numerical integral of (5) rather than by the GPD closed form of McNeil and Frey (2000), an earlier convention, leaves each comparison in place. At 2.5%, FZ0 is 1.849 against 1.851, the integral marginally better, DM 5.3, and the estimator-over-GARCH DM is 4.8 and 5.1 at the 1% and 2.5% levels. The splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$: DM 4.6–4.8).

Industry models as special cases. Equation (5) nests the standard toolkit, in the taxonomy of Kuester et al. (2006). Historical simulation sets $\widehat{\mu} \equiv 0$, $\widehat{\sigma}_t \equiv 1$, and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS) (Barone-Adesi et al., 1999; Hull and White, 1998) takes q as the unconditional empirical quantile of past *residuals*, i.e. Stage 2 with the state dependence deleted. GARCH-t (Bollerslev, 1987) takes $q = t_\nu^{-1}$, a fixed parametric shape. Whatever FHS captures, (3) can represent, and it adds conditional shape. CAViaR (Engle and Manganelli, 2004) sits outside the nesting. It autoregresses on the quantile directly, with no scale/shape split, and is the benchmark with which the estimator ties (Section 5).

Algorithm 1 (residual-hybrid estimator). (i) Fit (2) per asset on the estimation window. Store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names. Train the shape model by (3). (iii) Fit the GPD tail (4) by maximum likelihood on the pooled training excesses. Take the minimum with the body at $\tau \le p_0$ and rearrange. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data (overlay only; $c_\tau \equiv 0$ in the accuracy layer). (v) Forecast by (5). In *ongoing use* all stages refit on an annual walk-forward schedule. The paper's *evaluated* protocol freezes each stage once per split, the more conservative test since no stage ever adapts to the test era. Section 3.1 verifies the frontier separately under the true annual-refit schedule.

2.3 Estimating the shape: gradient boosting and neural quantile generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3). We use trees for point accuracy and the network when draws are needed (Online Appendix Table OA.3).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise by gradient boosting (Friedman, 2001), where $\eta = 0.06$ is a fixed shrinkage rate, $M = 250$ trees of depth three are grown, and h_m is the m-th regression tree, fitted by least squares to the pinball pseudo-residuals. Quantile regression forests (Meinshausen, 2006) are the other tree-based route and are not used here. Each new tree h_m is fit to the negative gradient of the pinball loss, which equals τ_k where the current model under-predicts and $\tau_k - 1$ where it over-predicts. The fitted grid is made non-crossing by monotone rearrangement, which sorts the values $\{q(\tau_1|s), \ldots, q(\tau_K|s)\}$ in τ and is weakly closer to the true monotone quantile curve in every L^p norm (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest point estimator, about 0.75% better pinball than the tuned network, consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN), the network at the core of GBC. A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s,\tau) \;=\; g\big(\psi(s)\circ c(\tau)\big), \qquad c_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=1}^{n_c}\cos(\pi i\tau)\,w_{ij}+b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication, $c(\tau)$ is a cosine embedding of the level with $n_c = 64$ features, and g, ψ are two-layer ReLU feed-forward nets of width 192 (128 in the untuned run) acting on the level embedding and on the standardized state respectively. The tuned variant uses tail-aware τ-sampling, monotone rearrangement of the τ-curve, and a conformal shift. The tail-aware sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, was set once, before the tuned run, so that about a tenth of the training levels fall below 2.5% against 2.5% under uniform sampling; the mixture and the Beta parameters were not tuned on any test data. It answers the untuned run, whose 99% breach rate was 3.2% under uniform sampling, and the tuned IQN row of Table 4 is a design revision made after that run and evaluated on a later sub-window. The network's role is generation rather than point accuracy: with $\tau \sim U(0,1)$, $z^* = H_\phi(s,\tau)$ draws from the fitted conditional law by inverse transform. The trade-off between the two estimators is set out under *Estimator choice* below.

How training works. The network minimizes the empirical version of (3) by stochastic gradient descent, each example drawn at a fresh level $\tau_{jt} \sim \lambda$ each epoch, so one set of weights carries the entire quantile curve. Training runs for a fixed budget of 12,000 minibatch steps (Adam, batch 512, rate 10^{-3}; 6,000 in the untuned run) with no early stopping. Monotone rearrangement and the conformal shift are applied *after* training, and the whole pipeline is refit on the annual walk-forward schedule using only past data. Layer sizes, learning rates, and seeds are reported in the replication package.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names. In generative-Bayes terms, H is an amortized map evaluated at any new (s,τ) without per-asset refitting. It works, at cold-start in particular, because the mapping is estimated across names rather than within one. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns a single mapping from observable state to residual shape, estimated from millions of asset-days across names. For a newly listed asset, the estimated mapping can be evaluated from the available state variables without per-asset refitting. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically. For the same reason the explicit own-history blends add nothing: the state vector already carries the name's own recent information.

Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's own recent dynamics, realized vol above all. Removing it costs +1.0%, while removing all cross-sectional characteristics costs only +0.5%. Characteristics matter at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an affine own-history shrinkage of the amortized quantile toward the name's trailing location and scale, with weight age/(age+80), both hurt at each listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly.

Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 (benchmark tier) on the M5 competition data, outside finance altogether.

Estimator choice by forecasting objective. The two estimators minimize the same loss over different index sets. The trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level, while the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum. The appropriate estimator depends on whether the application requires fixed quantile levels or the full conditional quantile curve. At fixed regulatory levels (VaR/ES reporting, limits, capital) the trees are preferable: best point accuracy, no GPU, no sampling. Where the application consumes draws (scenario generation, portfolio aggregation by simulation), the network is preferable. It learns one continuous conditional quantile map, so draws come by inverse transform with no grid interpolation, at a measured cost of ∼0.75% in point accuracy. A GARCH-t or FHS model also simulates, from its fixed innovation law, whereas a tree grid requires monotone interpolation first. If both are needed, run both from the same pooled residual panel. They share each upstream and downstream stage of (5), so swapping one for the other changes a single component of the pipeline.

The amortized estimation loop is stated as Online Appendix Algorithm OA.1.

2.4 Formal results: why shape misspecification must show up in the loss

The nonparametric edge concentrates where the trailing residual-shape diagnostics are extreme, an empirical finding with a structural motivation in the pinball loss. Two elementary results, with proofs in the Online Appendix, supply it; they motivate looking where the shape diagnostics are extreme but do not prove that mk_{63} estimates the wedge, a link established empirically.

Lemma 1 (pinball regret identity (standard)). *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q, finite by the moment condition. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(v) - \tau\big)\, dv \;\geq\; 0, \tag{7}$$

with equality iff $F(u) = \tau$ for Lebesgue-almost each u between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that closed interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof. In the Online Appendix. □

The extra loss a misspecified forecaster pays is thus locally quadratic in its quantile error, and the frontier claim reduces to whether shape misspecification opens a quantile wedge at the levels that matter. For a fixed Student-t shape that is too thin (McNeil and Frey, 2000; Kuester et al., 2006), it does:

Proposition 2 (tail wedge under tail-shape misspecification (elementary))**.** *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. For each pair $\nu_2 > \nu_1 > 2$ there exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2]$ such that $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$ for all $\tau < \tau^*$. The fatter-tailed law thus has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}\left(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)\right)^2$ at each level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof. In the Online Appendix. □

The tail-wedge motivates but does not establish the empirical score-ordering result. We claim no monotonicity theorem at the regulatory levels: because the innovations are unit-variance, a heavier tail can carry the *less* extreme 2.5% quantile, with a pair-dependent crossover τ^* (0.018 for t_3 against the normal, 0.023 for t_5 against t_8), so 1% sits below it and 2.5% just above (Patton, 2020). The bound is a fixed-level statement rather than a tail-limit one: because $m = m(\tau)$ shrinks as $\tau \downarrow 0$, the certified regret $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ stays positive at each level but does not grow into the tail, and in fact vanishes there. We therefore treat the score-ordering as an empirical regularity, motivated by the tail-wedge mechanism rather than implied by it and documented across the design panel, the frozen holdout, the FRTB comparison, and the cross-market universes.

Lemma 2 (validity of the generative sampler (standard))**.** *If $\tau \mapsto \widetilde{Q}^z_t(\tau) = Q^z_t(\tau) + c_\tau$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0, 1)$, then $Z^* = \widetilde{Q}^z_t(\tau)$*

has quantile function exactly $\widetilde{Q}_t^z(\cdot)$.

Proof. In the Online Appendix. □

Because the flexible class contains FHS as the constant-in-state special case (Barone-Adesi et al., 1999), empirical risk minimization attains in-sample pinball risk no worse than FHS, and the out-of-sample comparisons of Section 5 test whether that dominance survives estimation noise; CAViaR (Engle and Manganelli, 2004), outside this nesting, is the one rival the estimator only ties.

2.5 Comparison with generative Bayesian computation

Relative to the generative Bayesian computation of Polson and Sokolov (2023), the downside-risk target changes the estimator in five ways. The sampling measure is tail-weighted, $\lambda = \frac{1}{2}U(0,1) + \frac{1}{2}\mathrm{Beta}(0.3, 0.3)$ in place of $\tau \sim U(0,1)$, because uniform sampling starves the extreme levels that downside risk depends on. The object modeled is the GARCH-standardized residual rather than the raw return, so that the conditional-scale dynamics the parametric model already estimates well are kept. Beyond the training support, extrapolation is handed to the GPD tail (4) spliced at p_0, so that it is governed by extreme-value theory and not by the network. The conformal stage adds the exchangeability-based coverage statement of Proposition 1. Finally, the amortization runs across the real cross-section of hundreds of assets instead of across simulations of a single model, which is what produces the cold-start forecasts and transfer results of Section 2.3. The question changes as well. When such machinery improves on the parametric filter is not something generative computation addresses, and the score of the next subsection is built to answer it.

2.6 The misspecification score

We use "misspecification score" as shorthand for a diagnostic of local residual-shape stress under the fitted GARCH-t benchmark. It is a trailing-window moment monitor in the spirit of the PITbased density-forecast diagnostics of Diebold et al. (1998) and Berkowitz (2001), used as a rank rather than as a formal specification test.

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape (Bollerslev, 1987). Static heavy tails do not violate this, since ν absorbs them. A locally wrong shape does: recent residuals far fatter-tailed or more asymmetric than any fixed t, the time variation in conditional skewness and kurtosis that Hansen (1994) models parametrically. The score is therefore computed per asset-day over the trailing window $W_t = \{t-63, \ldots, t-1\}$ of $n_w = 63$ residuals, with window mean $\bar{z}$ and standard deviation s_z. The window *ends the day before the forecast target*, so each signal is lagged at least one day relative to the loss it is paired with. The frontier sorts of Section 4 pair S_{t-1} with the day-t loss, an alignment enforced by construction. The two moment statistics are

$$\mathrm{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \mathrm{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

In estimation, mk_{63} and sk_{63} are computed as the bias-corrected sample excess kurtosis and skewness. At the full window length the bias correction is a common affine rescaling of (9), so the pooled decile ranks that the frontier uses are unchanged. *Skewness* is the signed third moment sk_{63}. *Asymmetry* always means its absolute value $|\mathrm{sk}_{63}|$, departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-5,\ldots,t-1\}} |\widehat{z}_i|$. The frontier orders the edge by the decile of each of these signals. The composite score is the maximum of their trailing percentiles, $\mathrm{score}_t = \max\{\widehat{F}_t(\mathrm{mk}_{63}), \widehat{F}_t(|\mathrm{sk}_{63}|), \widehat{F}_t(J_5)\}$, where $\widehat{F}_t$ is each signal's percentile in the pooled test panel for the sorts of Table 1 and Figure 1, and in the trailing panel available before t for the expanding-cutoff check reported beside them. The composite is itself deciled. All three are predetermined (trailing-window) and computable on any asset with a fitted GARCH, so the frontier can be monitored in real time. Online Appendix Table OA.3 reports the associated per-decile actions.

Why these statistics. Conditional on the fitted location-scale specification (μ_t, σ_t), the distributional restriction the flexible family relaxes is the fixed t_ν innovation *shape*. The lowest-order ways a fixed symmetric law can be locally wrong are the third and fourth standardized moments (Bollerslev, 1987; Hansen, 1994). The statistic sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies. An exact t_ν sample has known positive excess kurtosis, but the score is used as a rank, so only its cross-sectional and temporal variation matters, not its t_ν baseline. J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent, and Section 4 shows that the realized edge concentrates there.

The monitoring procedure is stated as Online Appendix Algorithm OA.2.

3 Data and evaluation protocol

Equity panels come from CRSP (the Center for Research in Security Prices) daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations. Each modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: the top 200 common shares (share codes 10 and 11) by average market capitalization among those with at least 3,000 observations in the window. Returns are in percent and include CRSP delisting returns where present. A name whose delisting return CRSP does not record ends at its last regular return, an omission Shumway (1997) shows to be large and negative for performance-related delistings. The one delisting event in the design panel moves the main results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data.

The replication repository carries code, configuration, and derived statistics, and the panels can be rebuilt from the queries above by any licensed subscriber. The five evaluation panels are a 2014–2024 design equity panel (200 CRSP names, 221,600 test observations) for the frontier and estimator, a frozen-spec 2000–2013 holdout (200 names, 280,608 observations), a 140-name exception-restatement panel, and Bloomberg cross-asset (43) and country-index (26) sweeps. The Online Appendix tabulates them.

Splits are chronological within each name. The first 60% of a name's history is the estimation window and the last 40% is test. The final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH-t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on the pooled training-window residuals (the estimation window less the calibration split), the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule's survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the note that closes this section. Distributional accuracy is scored by mean pinball loss across a fixed grid of quantile levels, twelve in the FRTB comparison of Table 3 and eleven in the frontier sorts, a consistent scoring function for the quantile vector that is the point-mass case of the quantile-weighted CRPS (Gneiting and Ranjan, 2011), and joint (VaR, ES) pairs by the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019). Loss comparisons use per-date Diebold–Mariano statistics (Diebold and Mariano, 1995) with a Newey–West variance (Newey and West, 1987), and multi-model fields use the Model Confidence Set (Hansen et al., 2011). Exception behavior is tested by the Kupiec unconditional-coverage, Christoffersen conditional-coverage, and date-clustered breach tests at the two regulatory levels.

"Frozen-specification" has a specific meaning here: a run is frozen-specification when the full specification (signals, features, hyperparameters, split rule, and test statistics) is fixed and the predictions are written down before the data are pulled. We do not use the word *registered*, since no third-party registry or timestamp exists for them. The holdout pipeline is line-for-line the design-era pipeline, and both are public in the replication repository, so the two can be compared directly to confirm that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 4 ran under this rule. As a post-hoc check, the composite score also replicates on this holdout. Its top decile carries +1.12% (DM 2.51), marginally above its kurtosis component (+1.04%, DM 2.16), and +0.71% (DM 2.08) under the expanding prior-only cutoff. Because the composite was not the rule frozen at the time, this is replication on an already-examined sample, not a frozen-specification test.

3.1 Common shocks, calendar overlap, and the pooled learner

Two nested flexible specifications appear in the paper. The residual-hybrid holds the GARCH conditional scale fixed and models the standardized residuals; it produces the design-panel frontier (Section 4) and every VaR and ES forecast. A simpler returns-space variant, plain enough to fix in advance, carries the pre-committed out-of-sample and robustness checks. Because a 60/40 per-name split can place one name's training window on another's test dates (López de Prado, 2018), a pooled learner could meet a systemic event in training that a per-name model meets only out of sample. A strict calendar-time split closes that channel (Gu et al., 2020): each stage is re-estimated strictly before 2020-01-01 and tested after, and the top-decile edge survives (+2.47%, DM 6.22). It also survives a pre-2007 cut that removes the 2008 crisis from every training window (+0.29% overall, DM 3.08), and an annual walk-forward that refits the GARCH, residuals, score, and learner at each January-1 cutoff, under which the top decile still carries +2.12% at DM 4.41 while the average edge washes out (−0.22%, DM −0.59); the returns-space learner of the frozen holdout specification gives +2.47% (DM 6.05) and +0.32% (DM 0.8) under the same schedule. Stale benchmark parameters therefore do not produce the conditional result. The overlap mechanism and the full calendar-split, pre-2007, and walk-forward tables are in the Online Appendix.

3.2 Cross-sectional dependence, family-wise error, and the universe rule

Each main test collapses the cross-section to one number per date before any variance is estimated, and the bootstraps resample dates rather than asset-days, following the pooled-average panel test of Qu et al. (2024). Over the full 3×10 signal-decile grid, Romano–Wolf step-down family-wise control (Romano and Wolf, 2005, 2016) leaves nine cells significant at 5%, all of them frontier cells and including the top deciles of all three signals; any cell that does not survive is reported as indistinguishable from zero wherever it is discussed. The exploratory single-instrument universes carry their own Holm adjustment over 93 comparisons (Holm, 1979). Survivorship is addressed by a point-in-time re-selection that fixes the universe at the top 200 names as of Q1-2000, applies no minimum-history filter, and rides each delisting return to the end: on the 164 names that clear the fitting floors the overall edge is +0.58% (DM 8.96) and the frozen top bucket carries +2.94% at DM 6.6, larger than in the survivor-tilted holdout. The date-bootstrap construction, the cell-by-cell adjustment, and the point-in-time details are in the Online Appendix.

3.3 Reporting and inference conventions

The reporting and inference conventions used in the empirical sections that follow are fixed here. Edges are percentage reductions in pinball loss, so larger is better for the named model. A "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. Ratios are model-over-benchmark loss ratios, so below 1 favors the model and $1 - \text{ratio}$ is the edge ($0.973 =$ a 2.7% win, $1.043 =$ a 4.3% loss).

DM is the Diebold–Mariano t-statistic on the loss difference. The forecasts are frozen objects, so the comparison is of forecasts rather than of estimated models, the use Diebold (2015) endorses. For the per-date DM, losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$. The orientation is stated at each use (benchmark minus the residual-hybrid in its main comparisons, so that positive favors the residual-hybrid). The test is then run on the resulting ∼1,100-date series with a Newey–West (lag-10) variance (Newey and West, 1987), so the effective sample is the number of dates, not the number of asset-days. The lag replaces the $h - 1 = 0$ truncation of Diebold and Mariano (1995) because misspecified one-step forecasts leave serially correlated loss differentials.

Reported p-values are one-sided in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%), a direction fixed before the run for the residual-hybrid's main comparisons, and where a winner is instead chosen by the data the one-sided level reads as a two-sided level of twice its size. A t-statistic above 2.6 is significant under either convention, and marginal cases are flagged where they occur. A tie means the per-date DM cannot reject zero at the two-sided 10% level. The point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible we make it as one. For the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin, which is a two-one-sided-tests conclusion rather than a failure to reject.

The Model Confidence Set is computed on the same per-date loss matrix with the range statistic of Hansen et al. (2011) and a stationary bootstrap over dates (Politis and Romano, 1994) (mean block length 10 days, $B = 1{,}000$ replications in the main comparison, $B = 800$ in the CAViaR extension). Hansen et al. (2011) implement the set with a fixed-length block bootstrap (Künsch, 1989), and the stationary variant is the one used in the SPA test of Hansen (2005). Coverage numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism). For credible-interval coverage, closer to nominal is better in both directions. Correlations quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

4 The misspecification frontier

4.1 The within-universe result

On 200 CRSP names across 1,108 test dates ($\approx$221.6k asset-days), we bucket the per-day pinball edge of the residual-hybrid over GARCH-t into deciles of each misspecification signal. Both forecasters share the GARCH conditional scale, so any advantage enters through the residual quantile. Because that quantile is conditioned on trailing realized volatility as well as on the score, it can also act as a state-dependent correction to the daily scale, and Section 5 shows that on large capitalizations a realized-variance scale absorbs much of this component (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005). The frontier therefore measures the value of a state-conditioned residual law given a daily parametric scale, not a pure innovation-shape effect. Sorting forecast performance by a predetermined state is a conditional predictive-ability comparison in the sense of Giacomini and White (2006). Table 1 reports the sort on which the rest of the paper builds.

[Table 1 about here.]

The edge is concentrated in the top decile: the top mk_{63} decile carries +2.98% at per-date DM 10.5, the overall edge is +0.35% at DM 5.35, and the lower deciles are statistically indistinguishable from zero. The sort has a regression form. Taking the asset-day loss differential (GARCH-t pinball minus engine pinball) and the lagged score as the conditioning variable, the conditional-predictive-ability test of Giacomini and White (2006) with instruments (1, score) rejects equal conditional accuracy (Wald $\chi^2_2 = 73.6$ on the composite percentile, 90.0 on the kurtosis percentile), and the slope of the differential on the score is positive with a date-clustered t of 8.6 (composite) and 9.8 (kurtosis). On a date-by-date reading the engine has the lower loss on 74% of dates in the top decile and 55% in deciles one through nine.

GARCH-EVT on the same rows. The frontier is a statement about a state-conditioned residual shape, so the natural question is whether a conventional unconditional tail captures it. On the rows of Table 1, a McNeil–Frey GARCH-EVT forecast (the same GARCH-t filter, GPD tails fitted by maximum likelihood to the 10% most extreme training residuals per tail, empirical residual quantiles in between) shows no frontier. Against GARCH-t its top-decile edge is +0.31% (DM 3.6) per name and −0.16% (DM −2.7) with a pooled tail, and its overall edge is negative in both forms. The estimator beats the pooled GARCH-EVT by +2.64% (DM 8.9) in the top kurtosis decile and by +0.10% (DM 2.5) in deciles one through nine, and the per-name GARCH-EVT by +2.18% (DM 9.6) and +0.21% (DM 4.6). The same-rows table and the GPD threshold diagnostics are in the Online Appendix, together with the rest of the comparison set on the same rows (Table OA.10). Of the standard benchmarks, only SAV-CAViaR shows a top-decile edge over GARCH-t above one percent (+2.81%, DM 8.2). It models the quantile directly and reacts to $|r_{t-1}|$, so it carries the same frontier the engine does, and the two tie in every region (top decile −0.33%, DM

-1.8; overall -0.12%, DM -1.6, engine over CAViaR), the tie Section 5 reports on the FRTB battery. The conditional-predictive-ability slope of each benchmark's loss differential against the engine on the lagged score is positive with $t > 2$ for GARCH-t, per-name GARCH-EVT, pooled-tail GARCH-EVT, pooled FHS, per-name FHS, rolling FHS, EWMA and GJR-GARCH-skew-t, so the engine's advantage over each of them grows with the score; it is flat for historical simulation and SAV-CAViaR.

The composite score. Table 1 sorts on each signal separately. The monitor the paper uses is the composite score, the maximum across the three signals' trailing-panel percentiles (kurtosis, asymmetry, jump). Evaluated directly, its top decile is a 10% cell carrying +2.79% (DM 8.9), and the lower nine deciles sit between -0.3% and +0.35%. The edge survives the strictly non-anticipating expanding cutoff at +2.72% (DM 9.0). The three components each carry a top-decile edge on their own: +2.98%, +2.78%, and +1.96% (kurtosis, asymmetry, jumps). The maximum of decile ranks instead flags the union of the three top deciles, a 19% cell at +1.72%, so we report the re-deciled composite.

The score's content could reflect model-relative misspecification, departure from each name's own fitted t_{ν_i}, or the absolute level of recent tail activity. Converting each mk_{63} into a percentile of the simulated null of 63-window kurtosis under the name's fitted t_{ν_i} leaves the frontier essentially unchanged (top decile +3.0%, DM 8.5, against +2.98%, DM 10.5 raw). The normalization is nearly rank-preserving, however (Spearman 0.97, 82% top-decile overlap), so it cannot by itself separate the two.

A discriminating test instead favors the raw-tail-activity reading. In a within-date cross-sectional (Fama–MacBeth) regression of the per-observation edge on the raw score and the ν-relative percentile, the raw score carries the predictive content ($t = 7.8$), while the ν-relative percentile has significantly negative incremental content ($t = -5.2$). Double-sorting agrees. The top-raw-kurtosis-quintile edge falls from +1.9% for the most fat-tailed (low-ν) names to +1.5% for the thinnest-tailed, and observations high in the ν-relative score but not in raw kurtosis carry no edge (+0.3%, DM 0.5). The signal is thus the absolute level of recent tail activity rather than a per-name ν-normalized surprise. The signal still measures shape stress relative to the fitted model, since mk_{63} is computed on the standardized residuals the model treats as stationary t_ν. Elevated values flag states in which the realized residual shape is heavier or more asymmetric than the fitted law typically produces. A correctly specified low-ν t_ν can itself generate high realized kurtosis, which is why the score is treated as an indicator of stress and makes no claim to be a specification test (Section 2.6). The data reject the finer per-name normalization. A plausible reason is that a name's fitted ν is estimated from the same residuals and a low value is itself a symptom of the tail activity that matters, so dividing it out removes part of the signal.

Scale versus shape. The top decile follows large standardized residuals, and since a GARCH(1,1) inflates its conditional variance after such a shock, part of the edge reflects the standard filter's post-outlier scale error rather than any error in the innovation shape. The two can be separated with a jump-robust GARCH, which bounds the squared-innovation news term at $K\sigma_{t-1}^2$ (a $\sqrt{K}$-sigma cap that limits how far a single shock propagates into the variance) and so responds to a single shock only within that cap. Against that benchmark the top-decile edge falls from +2.98% (DM 10.5) to +0.45% (DM 4.7) at a three-sigma cap and to +0.27% (DM 2.5) at four sigma, while the robust filter alone accounts for +3.17% (DM 8.2) of the standard-GARCH gap in that decile. Most of the frontier is therefore daily-scale misspecification, which a robust scale corrects on its own; what remains under a robust scale is a conditional-shape edge of the order of a third to a half of a percent, smaller but still significant. The large-cap realized-variance decomposition of

Section 5 tells the same story. It is the reason the score is read as a monitor of parametric-model stress, part scale and part shape, and not as a diagnostic of shape alone.

Robustness of the frontier. Four checks in the Online Appendix address the main potential artifacts. The one-day score lag is not a look-ahead window: it gives the largest top-decile edge, and a five-day-old score keeps most of it (Online Appendix Figure OA.1). Re-forming the decile cutoff from strictly earlier dates, the rule the monitor uses, leaves 86% of top-decile membership unchanged and the edge intact (+2.45%, DM 8.3), so the concentration is not an artifact of the pooled-panel sort. A per-name FHS reweighted toward days of similar mk_{63} is significantly worse than plain FHS (per-date DM -11.3), so a parametric-adjacent fix cannot absorb the edge and the pooled learner is doing the work. A double sort on mk_{63} and trailing volatility shows the edge needs both a shape break and an active regime, peaking at +2.33% where both are present (Online Appendix Table OA.2). A superior-predictive-ability (SPA) test (Hansen, 2005) against the industry benchmarks returns $T^{\mathrm{SPA}} = 0$, and with the design-era threshold held fixed the top-decile edge is positive in every test year from 2020 to 2024.

The untouched-era holdout. Each design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze a simpler returns-space pipeline in full (signals, features, hyperparameters, split rule, and test statistics) and wrote down three predictions in advance: a positive and significant top-decile edge with a bottom decile near zero, a small positive overall edge, and a threshold-shaped decile profile matching Table 1. We then pulled CRSP daily returns for 2000–2013, an era not used at any earlier stage of the analysis. The pre-specified kurtosis-component frontier, the variant frozen in advance, replicates: the top-decile edge is +0.89% with per-date DM 2.16 ($p = 0.016$, 1,465 dates), deciles one through nine sit between -0.18% and $+0.28\%$, and the overall edge is +0.02%. The residual-hybrid composite replicates as well (Figure 1): its top decile carries +1.12% at per-date DM 2.51. This is the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. On the same holdout rows the pooled GARCH-EVT again shows no frontier (−0.15% against GARCH-t in the top decile, DM −2.8), and the frozen learner beats it there by +1.16% (DM 2.5), within noise elsewhere. The decile boundaries above are within-sample ranks of the holdout era, a partition that never touches a forecast but could not have been applied on day one. Two day-one-applicable rules recover the result: freezing the design-era thresholds as constants keeps the threshold shape (bins eight to ten at +0.82 to +1.02%, though the cell thins and the test loses power), and an expanding pooled 90th-percentile threshold over strictly earlier dates recovers the sort's edge with significance (+0.99%, DM 2.15). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge, and the frozen thresholds carry +2.94% at DM 6.6 on the point-in-time universe of Section 3.2.

[Figure 1 about here.]

The overall signal–edge correlation is low (0.05–0.10) because the relationship is a threshold nonlinearity, so the top-decile Diebold–Mariano statistics are the appropriate summary.

4.2 The frontier across universes

[Table 2 about here.]

Table 2 shows the same axis at work across markets. The decisive standalone wins are the hyperinflation and capital-control currencies, the Argentine peso and the Egyptian pound, 12–14% better and individually significant, whose residual kurtosis (∼2000–3300) sits two orders of magnitude above anything GARCH-t was built to absorb. Twenty-six country indices trace a gradient that tracks residual kurtosis rather than the development label, and forty-three cross-asset instruments extend the pattern into rates, commodities, and credit. The Korea-2026 crash is the negative control: a price crash under circuit breakers censors the return but leaves residual kurtosis ordinary, and GARCH-t correctly wins all twenty-three assets, so the frontier tracks residual shape and not price turbulence. The instrument-level entries are exploratory, and under a Holm step-down over the 93 comparisons (Online Appendix Table OA.4) seven survive. The aggregate crisis-tier FX statistic (DM 4.12) and the cross-universe kurtosis–edge gradients are the evidence this section rests on. A frequency dimension is left to future work: the five daily crypto series show no edge, while at hourly frequency the picture inverts.

4.3 Regime dependence and model selection

In the amortized panel the absolute edge grows about fourfold with turbulence, from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent, and both nonparametric estimators beat GARCH in every quintile. Gating, running GARCH on calm days and the nonparametric model when the score is high, does not improve on continuous use of the flexible model: a classifier gate routes 90.5% of days to the nonparametric model and matches always-nonparametric exactly, and a regression gate on the edge magnitude (Giacomini and White, 2006) still converges to it. This is consistent with Table 1: the conditional mean of the loss differential is positive where the score is high and indistinguishable from zero elsewhere, so no state pays to select the parametric quantile, and the per-day oracle gap (4.4%) is largely unforecastable from prior-day state. The amortized setting therefore uses the nonparametric model throughout; the residual-hybrid is the choice where forecast accuracy no worse than GARCH-t in the bulk is wanted, with the score as a monitor (Section 6).

5 VaR and Expected Shortfall forecast evaluation

5.1 Head-to-head accuracy and significance

Table 3 reports the full comparison on 140 CRSP names (155k observations, 1,108 dates).

[Table 3 about here.]

The stress-era replication. The comparison set was re-run on the pre-committed 2000–2013 holdout panel. Splits follow Section 3, so each name's test window runs from roughly mid-2008 through 2013, the crisis and its aftermath. The EVT-tailed residual-hybrid passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t-statistics of 0.17 at 99% and 0.35 at 97.5%, consistent with nominal at both regulatory levels through the crisis window. Historical simulation, by far the most common method in bank disclosures (Pérignon and Smith, 2010), and EWMA do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH-t is also well calibrated in this era (94% pass rate), as the frontier predicts. Calibration parity is the norm, and the residual-hybrid's added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative on the 2000–2013 holdout (breach 2.21% against the 2.5% target, $t = -1.09$, not significant). The safety margin widens when the calibration era is calmer than the test era, and the error runs in the direction regulators prefer.

Non-core comparisons. Table 4 adds two entrants from outside banks' standard toolkit. SAV-CAViaR, a leading semiparametric academic benchmark, ties the residual-hybrid, and the 90% Model Confidence Set on that run contains exactly these two, so the residual-hybrid ties the strongest non-nested academic rival and significantly beats every model in banks' standard toolkit.

The two tied models differ in what they deliver: SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history, so it cannot be applied to new listings; the residual-hybrid produces the full curve and extends to new listings. Pooled exception rejections are not counted against CAViaR, because pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, the residual-hybrid included; the date-clustered restatement is the appropriate reading. A neural IQN used as the residual model loses to trees at both tunings (Table 4): the tail-aware, monotone, conformally recalibrated version narrows the untuned gap by an order of magnitude and repairs the thin tail, but the remaining deficit is still significant, so trees are the stronger tabular estimator and the residual-hybrid's edge does not rest on the choice of shape learner. Two textbook FHS variants, per-name train-constant and per-name rolling 500-day trailing, were run on the identical panel; both lose to the residual-hybrid (DM 5.2 and 5.7, pinball 0.3562 and 0.3569 against 0.3551) and bracket the pooled variant used in Table 3.

[Table 4 about here.]

The ES columns of Table 3 are diagnostic only. Under the exact tail integral, each fat-tailed entrant's predicted ES exceeds the realized mean below its own VaR, by 9% (raw residual-hybrid, rolling FHS) to 22% (pooled FHS). The comparator conditions on each model's own breach set. The joint FZ0 score below is the ES ranking used in this paper.

The joint $(\mathrm{VaR}, \mathrm{ES})$ **score.** Because ES is not elicitable alone but the pair $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the comparison set was also scored under the FZ0 joint score

$$L_{\mathrm{FZ0},\alpha}(r, q, e) = -\frac{1}{\alpha e}\,\mathbb{I}\{r \leq q\}(q - r) + \frac{q}{e} + \log(-e) - 1, \qquad e \leq q < 0, \tag{10}$$

the 0-homogeneous member of the class (Patton et al., 2019). Lower values are better. On the large-cap sandbox subset used to fix the score's orientation, a Gaussian-innovation GARCH is significantly worst at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, Gaussian innovations are clearly inadequate. Among the fat-tailed entrants the FZ0 differences on large caps are small ($|\mathrm{DM}| < 1.4$), the frontier's own prediction that the shape edge is flat on well-behaved names. On a high-kurtosis subset the ordering favors the shape-aware forecaster, directionally but not significantly at that width. The pre-specified full-panel run, reported next, settles the magnitudes on the residual-hybrid's own forecasts.

[Figure 2 about here.]

On the full 200-name panel (221.6k test asset-days), VaR and ES are read from the coherent min-envelope curve Q_t^z of (5) and per-date DM inference is used throughout. The residual-hybrid's *accuracy layer*, the EVT-tailed variant with no conformal shift, attains the lowest FZ0 at both regulatory levels. At 1% it beats GARCH-t (DM 4.8) and FHS (DM 4.9), the two standard benchmarks the accuracy claim rests on. At 2.5% it again wins over GARCH-t (DM 5.1). On the same rows it also beats standalone GARCH-EVT, whose ES is the McNeil–Frey closed form, with a pooled tail by DM 4.8 at 1% and 6.8 at 2.5%, and per name by DM 5.2 and 5.4, so the gain over the conventional EVT construction is not confined to the pinball frontier. The dynamic quantile test of Engle and Manganelli (2004) (four hit lags and the VaR) passes at the 5% level for 86% of names at 1% and 70% at 2.5% for the accuracy layer, against 84% and 68% for GARCH-t and 83% and 62% for the pooled GARCH-EVT; the overlay lifts the 2.5% rate to 84%. Because a ranking under one consistent score need not hold under another (Patton, 2020), the Online Appendix reports Murphy diagrams (Ehm et al., 2016) for the VaR forecasts: the accuracy layer has the lower elementary score against GARCH-t on 70% of the threshold grid at 1% and 65% at 2.5%, and against the pooled GARCH-EVT on 85% at both levels, losing in a band of thresholds near the typical VaR level. Extending the joint score to the whole comparison set on the same rows (Online Appendix Table OA.11), the accuracy layer has the lowest mean FZ0 at 1% and is within noise of GJR-GARCH-skew-t at both levels (DM 1.1 and -0.3) and of the Taylor model at 2.5% (DM -0.1); the 90% Model Confidence Set on the per-date FZ0 series contains the accuracy layer, the conformal overlay, the pooled body and GJR-GARCH-skew-t at 1% and the accuracy layer, the pooled body, GJR-GARCH-skew-t and the Taylor model at 2.5%, and excludes GARCH-t, every FHS variant, both GARCH-EVT forms, the GAS model, EWMA and historical simulation at both levels. On the eleven-level pinball the set contains the pooled body and SAV-CAViaR and excludes the accuracy layer, since the EVT branch costs pinball, as Table 3 reports for the twelve-level battery, and is kept for the exception tests it passes. The two closest dynamic $(\mathrm{VaR}, \mathrm{ES})$ benchmarks are the one-factor score-driven GAS of Patton et al. (2019) and the ES-CAViaR of Taylor (2019). Each is estimated per name on the same panel by minimizing the same FZ0 loss, and under a three-start optimizer both converge for all 200 names, so the single-start instability of an earlier attempt does not arise. The accuracy layer beats the GAS at both levels (DM 9.5 at 1%, 8.6 at 2.5%), and it beats the Taylor model at 1% (DM 3.6) while tying it at 2.5% (DM 0.1), the near-tie the fat-tailed benchmarks all show at 2.5% where the shape edge is thin.

This ranking survives the annual walk-forward refit of Section 3.1: re-estimating every stage at each January-1 cutoff and re-scoring the 2020–2024 panel, the accuracy layer keeps the lowest FZ0 at both levels, beating GARCH-t and FHS by DM 3.6 and 2.7 at 1% and 3.1 and 2.6 at 2.5% (one-sided $p < 0.005$). Refitting narrows these margins without closing them, unlike the average pinball edge, which closes under the same refit.

The accuracy layer is thus one configuration that attains the lowest FZ0 at both levels and passes the aggregate date-clustered exception test at 99% and marginally at 97.5%. Per-name diagnostics show residual cross-sectional calibration heterogeneity (82% and 89% pass rates at

99%, Online Appendix Figure OA.2). The one failed test is the pooled 155k-observation Kupiec at 97.5%, which at this sample size rejects very small deviations. The frontier concentrates the joint-score edge as it does the pinball edge: the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.66% against the 2.5% target) and gives back the FZ0 advantage there. The static-overlay variant loses to GARCH-t on the 2.5% joint score (DM -1.8), and the concession is largest in the top misspecification decile, where the static shift's joint-score deficit reaches DM -7.1.

The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative (wider bands, fewer breaches than nominal). Where the conservative coverage margin is valued, the shift is kept; where FZ0 efficiency is the criterion, the EVT tail is run alone, and it passes the exception tests on its own (Online Appendix Figure OA.2).

Most of the concession reflects a frozen margin. The adaptive form of the shift (Gibbs and Candès, 2021), updated daily from the panel breach frequency, shrinks the margin out of era (from -0.35 to -0.10, breach 1.83% $\rightarrow$ 2.06%) and erases the overall concession. The adaptive variant ties GARCH-t on 2.5% FZ0 (DM -0.3). The top-decile cost survives the adaptive shift too, at -4.8 and -6.0 for the two update rates against the static shift's -7.1, the price of a margin where the bands are widest. The static shift also worsens per-name coverage (Kupiec$_{97.5}$ pass rate 48%, against 71% unshifted and 69% adaptive), because over-coverage fails the test from the other side. Each "lowest FZ0" statement refers to the accuracy layer.

Direct ES calibration backtests. Because exception counts test VaR and not ES, two direct ES calibration backtests, the McNeil–Frey exceedance-residual test (McNeil and Frey, 2000) and the Acerbi–Székely Z_2 statistic (Acerbi and Székely, 2014), complement the jointly consistent FZ0 loss above. Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a joint score. Both are computed on the residual-hybrid's own min-envelope forecasts over the same 200-name panel, with a stationary block bootstrap over the 1,108 test dates (Online Appendix). The Gaussian filter is worst on each ES metric, and among the fat-tailed entrants no single model dominates each diagnostic. The residual-hybrid has the Z_2 closest to zero at 1% and ties GARCH-t at 2.5%, and the residual-hybrid and GARCH-t are the two models that pass the 1% Kupiec test. On the McNeil–Frey residual, FHS, which targets the empirical tail directly, is closest to zero. The calibration tests show that the residual-hybrid's tail is well calibrated. The comparative result is the FZ0 loss, where the edge over GARCH-t and FHS is significant at both levels.

5.2 The ten-day horizon: a direct multi-day extension

The multi-day results come from a direct amortized model trained on h-day cumulative returns; the residual-hybrid of Algorithm 1 is a one-day architecture, and no ten-day claim attaches to it. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) its pinball edge over $\sqrt{h}$-scaled GARCH grows from $\sim$0.4% at $h = 1$ to $\sim$1.4% at $h = 10$ and holds in the 2020–2022 stress window (Online Appendix Figure OA.3). The ten-day tail advantage is real in that era but does not survive the pre-committed 2000–2013 rerun: out of era the two methods tie on accuracy (direct edge +0.49%, DM 0.07) and the calibration verdict reverses, with $\sqrt{h}$-scaled GARCH-t almost exactly calibrated through the crisis while the direct tail under-states it. In a 2008-type era the direct ten-day model should therefore be run with the $\sqrt{h}$-scaled parametric number alongside it as a conservative envelope. The ten-day evidence is therefore mixed; the full tables are in the Online Appendix.

5.3 Cross-asset evidence

Across the 43-instrument sweep the relationship is heterogeneous. The volatility indices (VIX, V2X, VVIX) and Baltic freight post the largest accuracy gains and survive the multiplicity adjustment (Online Appendix Table OA.4), but their calibration is open. VIX's breach-independence is marginal (one-sided $p = 0.041$), and credit and Baltic freight fail it, so all are reported as accuracy edges rather than calibrated wins. An apparent electricity win reversed once the return transformation was corrected for near-zero prices. Instrument-level results and diagnostics are collected in the Online Appendix.

The scale–shape decomposition. Against a Realized GARCH (Hansen et al., 2012) fitted on intraday data, the daily-data core concedes FZ0 at both levels (DM 6.2 and 4.5). Re-fitting the shape learner, EVT tail, and score on the realized residuals, however, leaves the score within noise. On large caps the frontier is therefore largely daily-scale misspecification. Where intraday data are unavailable or too sparse to build a realized measure (new listings, illiquid names, many cross-asset instruments), the daily score is the version of the diagnostic that can be computed (Online Appendix).

6 Extensions and implications

The full estimation and forecasting pipeline is stated in the Online Appendix.

Using the score. The score is available *ex ante*: a risk manager computes today's value from today's residuals and uses it for tomorrow's forecast. Its role is to stratify, not to switch models day by day, because the per-day oracle gap (4.4%) is largely unforecastable from prior-day state and a learned gate collapses to always-nonparametric (Section 4.3). The one setting in which it drives a switch is the per-name standalone model, where nonparametric quantiles underperform on calm days. As a monitor it is cheap, since its input is the trailing residual window the scale stage already produces, and it is a slow state that need not be watched daily. On capital, the residual-hybrid is a one-day single-name VaR/ES forecaster of the kind that feeds an internal model, not the ten-day portfolio-level stressed ES that sets the FRTB charge; the paper makes no capital-release claim, because the predicted-versus-realized ES wedge conditions on each model's own breach set and no defensible arithmetic converts it to capital (Basel Committee on Banking Supervision, 2019). The narrower implication is that the ES-based component of the charge scales with the ES forecast.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant here that can produce a conditional risk forecast for an asset with no return history. An IPO or newly listed ETF gets day-one quantiles from its characteristics and first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names; the residual-hybrid of Section 2.2 attaches once a scale filter is estimable. The amortized forecast beats own-history benchmarks at each listing age, and the edge is largest when the name is young (∼6–10% of pinball in the first trading month). The ablation in Section 2.3 traces the handover from characteristics to the name's own realized volatility as its history lengthens.

7 Conclusion

A fixed-shape parametric filter is hard to beat where its shape assumption holds, and for most assets on most days the flexible model cannot improve on it. The flexible model's advantage is concentrated in a small set of high-stress states identified by the real-time misspecification score. The decomposition attributes most of this advantage to post-outlier variance overstatement by the standard GARCH filter; a smaller but significant conditional-shape component remains under a jump-robust scale. An estimator that exploits both components, while showing no detectable accuracy loss relative to the benchmark elsewhere, improves on the accuracy of the market-risk models in standard use when judged by one-day diagnostics at the FRTB backtesting levels, at calibration parity with the best of them and short of the desk-level approval a full internal model must pass. Because the estimator pools its residual model across names, it also forecasts tail risk for assets with no history of their own, forecasting new listings from their first day as no per-asset model can.

These conclusions carry qualifications. The score is predictive rather than a necessary or sufficient condition: high residual kurtosis coincides with ties in some markets, and in credit and freight the accuracy edges sit alongside failed breach-independence tests. The shape component is the smaller part of the edge, since the decomposition attributes most of it to daily-scale misspecification that a jump-robust or realized-variance scale removes (Section 5), and what survives is real but modest. Where intraday data are too sparse to build a realized measure, the daily model and its score remain the available tool. And the score serves as a monitor rather than a day-to-day switching rule, since the day-ahead oracle gap is largely unforecastable (Section 4.3). Credit and freight retain accuracy gains but fail breach-independence tests, which makes dependence-aware calibration a priority for those applications. Distributional uncertainty around these VaR and ES forecasts and the multivariate portfolio tail are left to future work.

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Conflict of interest

The author declares no conflicts of interest.

Ethics statement

This study uses licensed secondary market data (CRSP/WRDS and Bloomberg) under the terms described in the data availability statement. It involves no human subjects and required no ethics approval.

Data and code availability

A public replication package at `github.com/sean-qin-usa/downside-risk-paper` covers every WRDS-based table and figure in this paper and preserves the Bloomberg-based exhibits as run from their derived series. It provides estimation code for all stages of the estimator ((2)–(5)), the misspecification score (9), the FRTB comparison, and the walk-forward schedule, seeds, and hyperparameter grids referenced in the text, together with a mapping from each table and figure to the script and result file that produce it. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author's institutional licenses and cannot be redistributed; the package includes the code to rebuild each panel from those sources, the derived non-proprietary series, and a synthetic example dataset on which the full pipeline runs end-to-end without licensed data. The holdout's frozen specification, including its written-in-advance predictions, is recorded in the package alongside the design-era pipeline so that the two can be compared line by line. A citable archival snapshot (Zenodo DOI) will be minted for the accepted version.

Notes

[1]The peaks-over-threshold formulation (Balkema and de Haan, 1974; Pickands, 1975) uses tail data more efficiently than block maxima (Embrechts et al., 1997) and is standard in risk management (McNeil and Frey, 2000). Here $\xi > 0$ is the heavy-tailed case.

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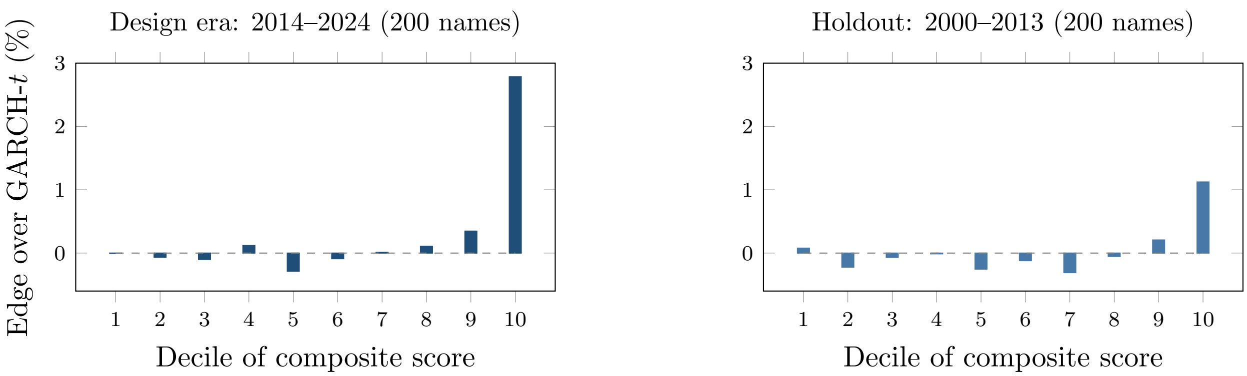

Figure 1: The frontier in and out of era, by decile of the composite misspecification score. Left: the design era (2014–2024). The top decile carries +2.79% at per-date DM 8.9, while deciles one through nine sit within ±0.35%. Right: the composite score run once on 2000–2013 CRSP data. The top decile carries +1.12% at per-date DM 2.51 over 1,465 dates. The threshold shape (edge concentrated in the top decile, near zero elsewhere) survives an untouched era containing the 2008 crisis.

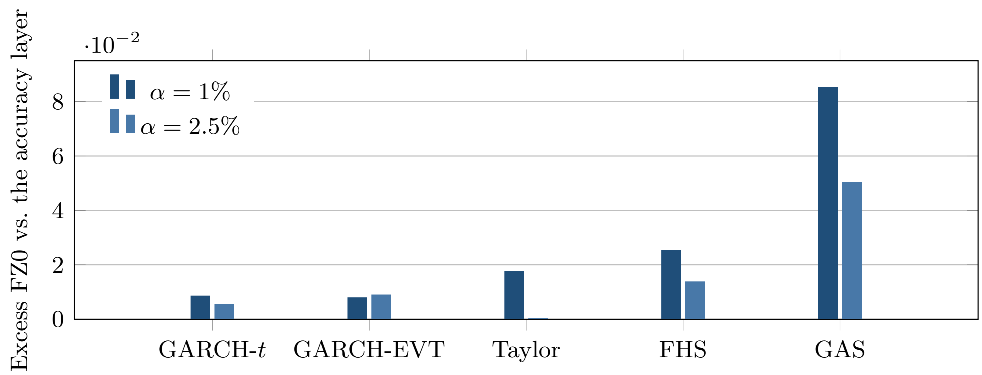

Figure 2: Joint (VaR, ES) FZ0 loss of each benchmark in excess of the residual-hybrid's accuracy layer (the EVT-tailed variant without the conformal shift), on the full 200-name panel. A positive bar means worse than the accuracy layer. The accuracy layer (the zero baseline) attains the lowest FZ0 at both regulatory levels, beating GARCH-t (DM 4.8 at 1%, 5.1 at 2.5%), pooled-tail GARCH-EVT (DM 4.8 and 6.8, same-rows run) and FHS (DM 4.9), together with the two FZ-estimated dynamic-ES benchmarks fitted per name by multi-start FZ0 minimization, the generalized autoregressive score (GAS) model of Patton et al. (2019) (DM 9.5 and 8.6) and the ES-CAViaR of Taylor (2019) (DM 3.6 at 1%, a tie at 2.5%). SAV-CAViaR, scored on its own common sample (Table 4), ties the accuracy layer and is omitted here; the conformal-overlay variant, which concedes FZ0 at 2.5%, is discussed in the text.

Table 1: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing misspecification score (200 CRSP names, 1,108 dates, 221.6k asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\lvert\mathrm{sk}_{63}\rvert$ (asymmetry)	jump_5
1–8 (bulk)	≈ 0	≈ 0	mixed
9	+0.46	+0.59	—
10 (top)	+**2.98**	+**2.78**	+1.96
Overall		+0.35	

Per-date Diebold–Mariano: top mk_{63} decile +2.98%, DM 10.5; $\lvert\mathrm{sk}_{63}\rvert$ +2.78%, DM 9.85; jump_5 +1.96%, DM 7.63 (all $p \approx 0$). Overall +0.35%, DM 5.35; the composite's bottom decile is indistinguishable from zero (−0.00%, DM 0.36). The composite score's top decile carries +2.79%, DM 8.91 (Figure 1). Top-decile signal means: kurtosis ≈ 25, $\lvert\mathrm{skew}\rvert \approx 4$. jump_5 is U-shaped, its lowest decile also carrying an edge (+1.17%, DM 5.5), so its bulk is mixed rather than near zero.

Table 2: The frontier across universes. Ratio = nonparam/GARCH-t pinball (< 1: nonparametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM −13.7
	EM (10)	1.013	8	GARCH, pooled DM −9.5
	crisis (7)	0.973	∼1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86, $p = .002$
	USDEGP	0.857	1920	DM 3.80, $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins[a]
Cross-asset (43)	vol indices	∼0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
13/43 significant wins overall, corr(log kurt, edge) = 0.43				
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM −2.5 to −5

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Exploratory: only Kenya survives the 93-comparison familywise adjustment (Online Appendix Table OA.4). Cross-index corr(log residual kurtosis, edge) = 0.53. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating that kurtosis is neither necessary nor sufficient.

Table 3: FRTB comparison: average pinball loss (twelve quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	$Breach_{99}$	$Kupiec_{99}$ p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	-6.37 / -5.83	1.09%	0.00
+ EVT tail	0.3555	-6.92 / -5.91	0.91%	0.00
GARCH(1,1)-t	0.3561	-6.69 / -5.65	1.06%	0.026
GJR-GARCH-skew-t	0.3562	-6.58 / -5.84	1.05%	0.062
GARCH-FHS (per-name)	0.3562	-6.61 / -5.84	1.11%	0.00
GARCH-FHS (pooled)	0.3566	-6.90 / -5.67	0.99%	0.79
GARCH-FHS (rolling 500d)	0.3569	-6.53 / -5.98	1.09%	0.00
EWMA (RiskMetrics)	0.3606	-5.52 / -5.63	1.63%	0.00
Historical simulation	0.3619	-7.21 / -6.86	0.84%	0.00

All columns come from a single common-sample run, provided in the replication package: the 140 CRSP names with the most complete history (at least 1,500 return observations each), over the common 1,108 test dates (155k asset-days), so each per-name exception test is adequately powered. The rolling-FHS burn-in then trims the same early dates from all models identically. Predicted ES is the numerical tail integral $\alpha^{-1}\int_0^\alpha Q(v)\,dv$: closed forms for the t and normal entrants, 200-node integration of the Hansen skew-t inverse CDF, and a matched 20-node midpoint rule for the empirical quantile models and for the residual-hybrid, whose VaR and ES are both read from one monotonized min-envelope curve Q_t^z of (5). A three-node tail-quantile average, a common shortcut, understates predicted ES by 2–6% relative to this integral. Diebold–Mariano vs the residual-hybrid: GARCH-t 5.0, GJR-skew-t 5.0, FHS pooled 6.2, per-name 5.2, rolling 5.7, HS 7.5, EWMA 8.7 (all $p \approx 0$, rounded, and stable across reruns). The EVT variant concedes 0.0004 pinball (DM $\approx$6) to the first row, the raw residual-hybrid without the tail, as the price of that tail. The 90% MCS contains the raw residual-hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a diagnostic, not a ranking: "realized" is the mean return below each model's own VaR, so the conditioning set is model-dependent, and under the numerical integral each fat-tailed entrant exceeds its own-tail comparator by 9–22%. The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew-t row: Hansen inverse CDF with the fitted degrees of freedom and skewness (median 4.2 and -0.015), validated against the symmetric-t limit and against an independent reference implementation.

Exception columns are pooled panel diagnostics: at 155k observations a breach-rate deviation of ± 0.1 points rejects in either direction (the EVT variant's 0.91% sits on the conservative side, pooled FHS happens to land at 0.99%), so pooled p-values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement (Online Appendix Figure OA.2): per asset the EVT-tailed residual-hybrid passes Kupiec at 99% for 82% of names and Christoffersen conditional coverage for 89% (at 97.5%, 71% and 79%), and the date-clustered test passes at 99% ($t = 0.93$) and marginally at 97.5% ($t = 1.85$) while the raw residual-hybrid is rejected ($t = 4.2$): the tail stages carry the regulatory calibration, and the 2008-era re-run confirms it out of era. At 97.5% the split-conformal recalibration is the stage that repairs pooled Kupiec (GBM-recal $p = 0.24$).

Table 4: Non-core comparisons. Each entrant is scored against the residual-hybrid on its own common sample, so loss levels are comparable within a row but not across rows.

Entrant	Loss	Residual-hybrid	DM	Breach_{99}	Verdict
SAV-CAViaR	0.3461	0.3460	0.4	—	tie ($p = 0.34$)
Neural IQN, untuned	0.3650	0.3551	13.3	3.2%	loses
Neural IQN, tuned[a]	0.3638	0.3632	3.03	0.92%	loses

[a] Tail-aware τ-sampling, monotone rearrangement, and a per-level conformal shift calibrated on the first half of the test window and scored on the second half; the residual-hybrid in this row is recalibrated the same way, so the loss levels are not comparable with Table 3.

DM is the per-date Diebold–Mariano t-statistic against the residual-hybrid; positive favors the residual-hybrid. The 90% Model Confidence Set contains the residual-hybrid alone on the Table 3 run, and the residual-hybrid together with CAViaR on the CAViaR run. Breach rates were not computed for the CAViaR run, which reports one quantile level per fit.